



ORIENT

Photo coupler

Product Data Sheet

MPN: ORPC-847 series of GK

Customer: _____

Date: _____

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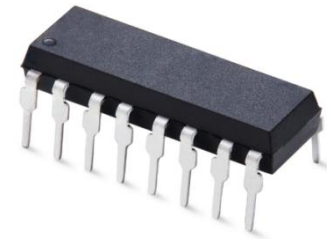
www.orient-opto.com

Preliminary

This datasheet is a preliminary design specification, and the formal specifications are subject to the recognition letter with jointly signed

1. Features

- (1) Current conversion ratio
(Min 50% Working condition $I_F=5\text{mA}$, $V_{CE}=5\text{V}$)
- (2) Insulation Voltage = 5,000Vrms,
- (3) Response Time
(tr: TYP. 2.9 μs ; tf: TYP. 4.5 μs at $V_{CE}=2\text{V}$, $I_C=2\text{mA}$, $R_L=100\ \Omega$)
- (4) ESD pass HBM 8000V/MM 2000V
- (5) Safety approval
UL approved (No.E323844)
VDE approved (No.40029733)
- (6) In compliance with RoHS, REACH standards
- (7) MSL Class I



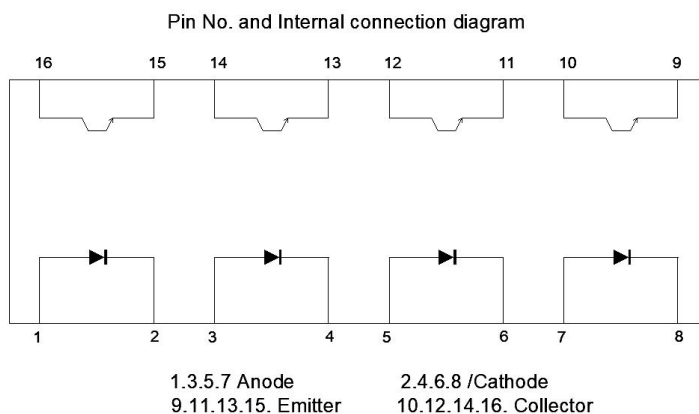
2. Instructions

- (1) ORPC-847-(GK) series optical coupler consists of four GaAs transmitting tubes and four NPN transistors
- (2) .Pin pitch of ORPC-847-(GK) is 2.54mm

3. Application Range

- (1) Switching power supply
- (2) Ammeter
- (3) Computer
- (4) Instrumental application, measurement machine
- (5) Imbursement equipments, duplicating machine, automat
- (6) Family-use electric equipments, such as fans
- (7) Signal transforming systems

4. Functional Diagram



5. Absolute Maximum Ratings (Ta=25°C)

Parameter		Symbol	Rated Value	Unit
Input	Forward Current	I_F	60	mA
	Peak forward current(1us pulse)	I_{FP}	1	A
	Reverse Voltage	V_R	6	V
	Consume Power	P	70	mW
Output	Collector and emitter Voltage	V_{CEO}	80	V
	Emitter and collector Voltage	V_{ECO}	7	
	Collector Current	I_C	50	mA
	Consume Power	P_C	150	mW
Total Consume Power		P_{tot}	200	mW
*1 Insulation Voltage		V_{iso}	5000	V _{rms}
Maximum transient isolation voltage (Insulating oil test)		V_{IOTM}	8000	V _{peak}
Maximum repetitive peak isolation voltage		V_{IORM}	850	V _{peak}
Operation Temperature		T_{opr}	-55 to + 125	°C
Storage Temperature		T_{stg}	-55 to + 150	
*2 Soldering Temperature		T_{sol}	260	

*1. AC Test, 1 minute, humidity = 40~60%

Isolation voltage shall be measured using the following method.

(1) Short between anode and cathode on the primary side and between collector and emitter on the secondary side.

(2) The isolation voltage tester with zero-cross circuit shall be used.

(3) The waveform of applied voltage shall be a sine wave.

*2. soldering time is 10 seconds

6. Electrical optical characteristics at TA=25°C

Parameter		Symbol	Min	Typ.*	Max	Unit	Condition
Input	Forward Voltage	V_F	---	1.2	1.4	V	$I_F=20\text{mA}$
	Reverse Current	I_R	---	---	5	μA	$V_R=5\text{V}$
	Collector capacitance	C_t	---	30	250	pF	$V=0, f=1\text{KHz}$
Output	Collector to emitter Current	I_{CEO}	---	---	100	nA	$V_{CE}=20\text{V}, I_F=0\text{mA}$
	Collector-Emitter Breakdown Voltage	BV_{CEO}	80	---	---	V	$I_C=0.1\text{mA}$ $I_F=0\text{mA}$
	Emitter-Collector Breakdown Voltage	BV_{ECO}	7	---	---	V	$I_E=0.01\text{mA}$ $I_F=0\text{mA}$
Transforming Characteristics	*1 Current conversion ratio	CTR	50	---	600	%	$I_F=5\text{mA}$ $V_{CE}=5\text{V}$
	Collector Current	I_C	2.5	---	30	mA	
	Collector and Emitter Saturation Voltage	$V_{CE(sat)}$	---	0.1	0.2	V	$I_F=20\text{mA}$ $I_C=1\text{mA}$
	Insulation Impedance	R_{iso}	5×10^{10}	1×10^{12}	---	Ω	DC500V 40~60%R.H.
	Floating Capacitance	C_f	---	0.6	1.0	pF	$V=0, f=1\text{MHz}$
	Transforming Frequency	f_c	---	80	---	kHz	$V_{CE}=5\text{V},$ $I_C=2\text{mA},$ $R_L=100\Omega,$ -3dB
	Rise Time	t_r	---	2.9	10	μs	$V_{CE}=2\text{V},$ $I_C=2\text{mA},$ $R_L=100\Omega$
	Fall Time	t_f	---	4.5	10	μs	

*1 Current Conversion Ratio = $I_C / I_F \times 100\%$



7. Rank table of current transfer ratio (CTR)

Grade Sign	Min (%)	Max (%)
NO BIN	50	600
BC Rank	130	400
CD Rank	200	600
CD1 Rank	300	500

Note: Working condition: $I_F=5\text{mA}$, $V_{CE}=5\text{V}$, $T_a=25^\circ\text{C}$

8. Order Information

Part Number

ORPC-847XV-W-Y-Z-(GK)

Note

X = Lead form option (S, M or none)

V = CTR Rank (BC, CD, CD1 or none)

W = Lead frame option (F: Iron, C:copper)

Y = 'V' code for VDE safety (This options is not necessary).

Z = 'G' code for Halogen free. (This options is not necessary).

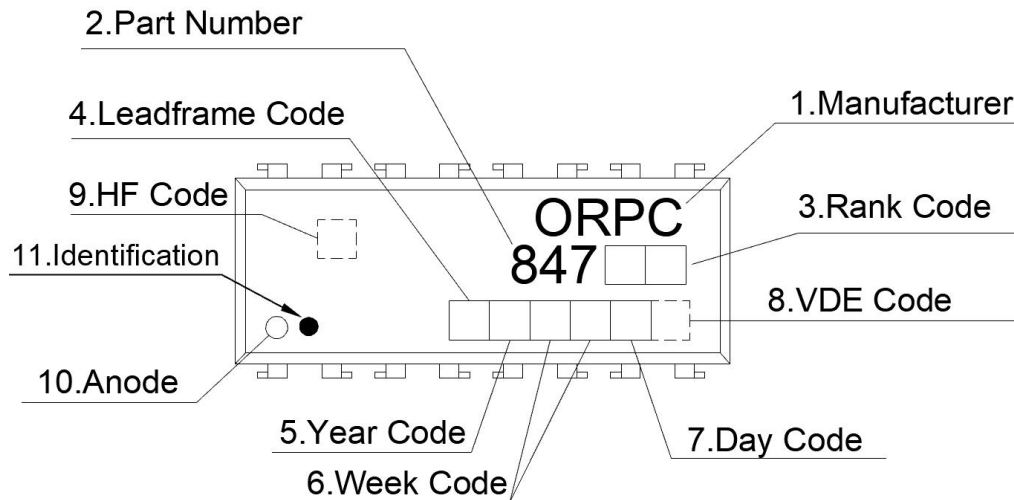
GK = Field code.

* Halogen Free can be selected.

* VDE Code can be selected.

Option	Description	Packing quantity
None	Standard DIP-16	24 units per tube
M	Wide lead bend (0.4 inch spacing)	24 units per tube
S	Surface mount lead form (low profile)	24 units per tube

9. Naming Rule



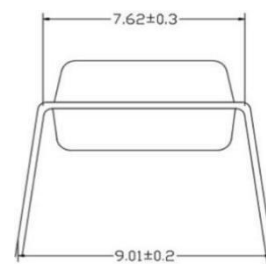
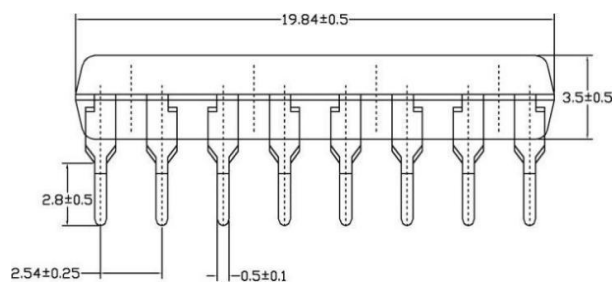
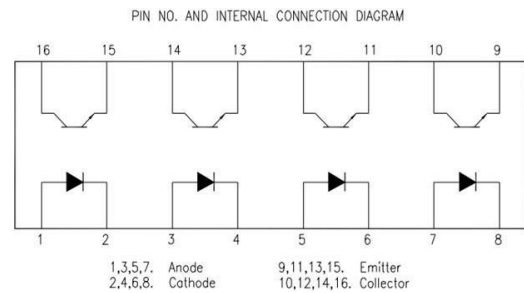
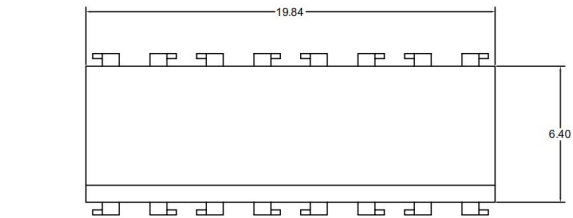
- (1) Manufacturer : ORIENT.
- (2) 847 denotes Part Number.
- (3) CTR Rand .
- (4) Lead frame Code : 'F' means Iron, 'C' means Copper.
- (5) Year Code : '1' means '2021' and so on.
- (6) Week Code : 01 means the first week, 02 means the second week and so on.
- (7) Day Code : "A to G" means "Monday to Sunday"
- (8) VDE Code . (Optional)
- (9) HF Code : Halogen Free. (Optional)
- (10) Anode.
- (11) Identification.

* Halogen Free Mark can be selected.

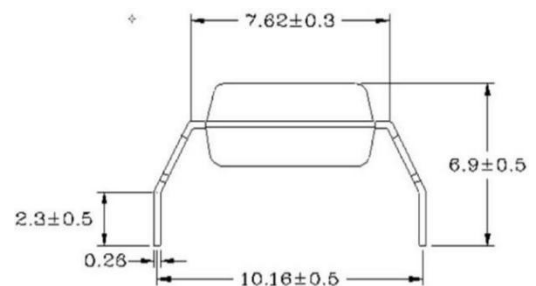
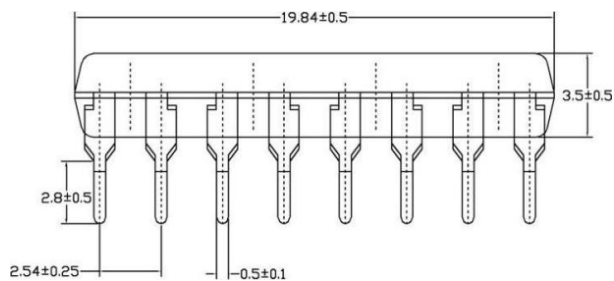
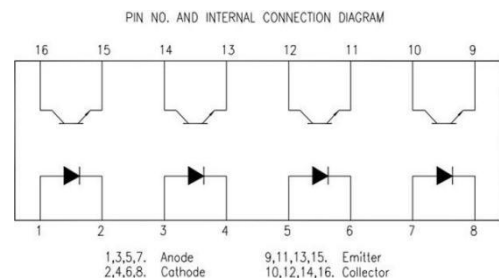
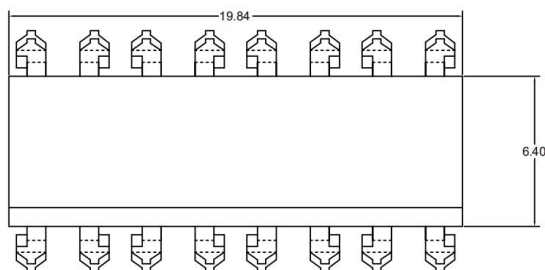
* VDE Mark can be selected.

10. Outer Dimension (Unit: mm)

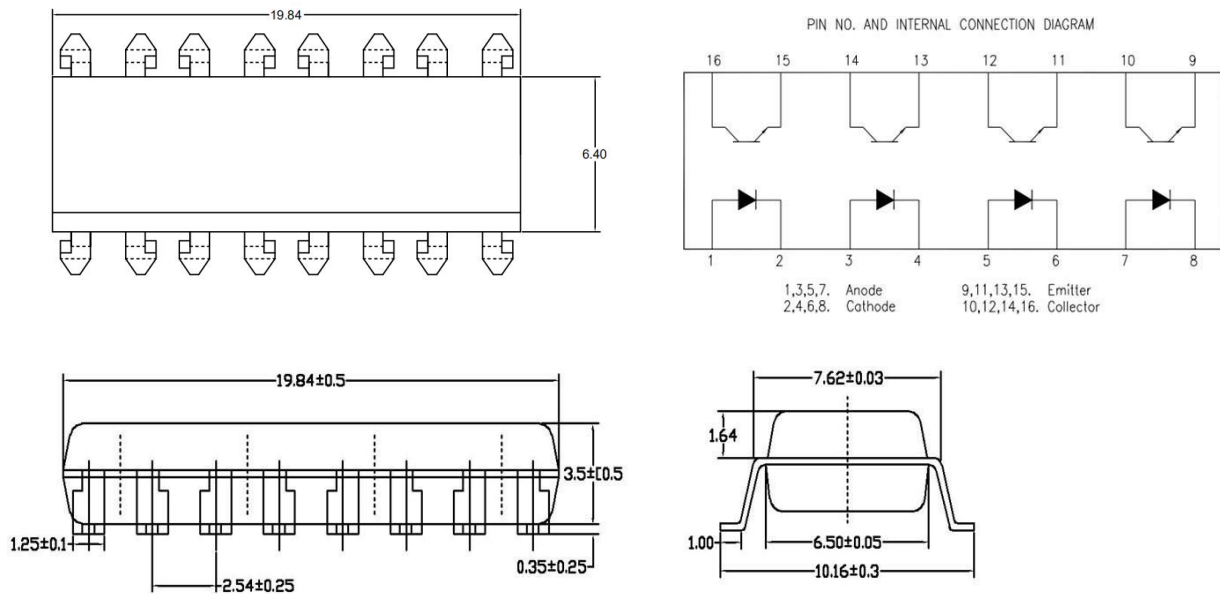
1. ORPC-847



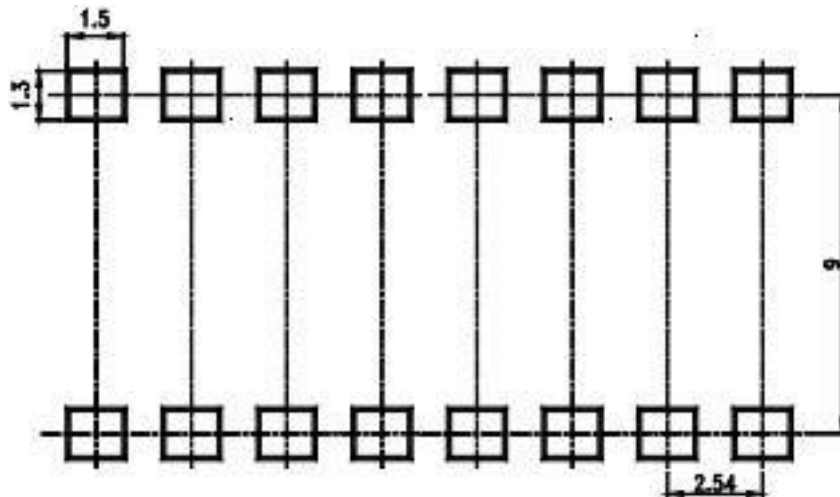
2. ORPC-847M



3. ORPC-847S



11. RECOMMENDED FOOT PRINT PATTERNS (MOUNT PAD)



Unit : mm

12. Package Dimension

(1) package dimension

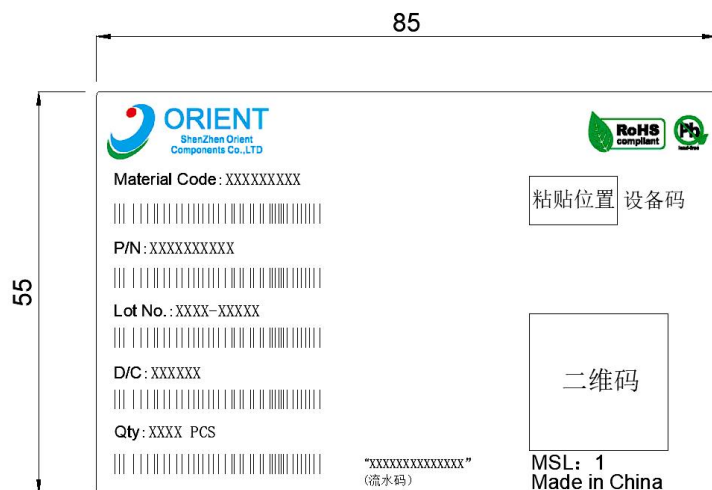
DIP Type

Packing Information	
Packing type	Tube
Qty per Tube	24pcs
Small box (Inner) Dimension	525*128*60mm
Large box (Outer) Dimension	545*290*335mm
The Amount per Inner Box	1,200pcs
The Amount per Outer Box	12,000pcs

SOP Type

Packing Information	
Packing type	Tube
Qty per Tube	24pcs
Small box (Inner) Dimension	525*128*60mm
Large box (Outer) Dimension	545*290*335mm
The Amount per Inner Box	1,000pcs
The Amount per Outer Box	10,000pcs

(2)Packing Label Sample



Note:

1. Material Code :Product ID.
2. P/N :Contents with "Order Information" in the specification.
3. Lot No. :Product weeks.
4. D/C :Product data.
5. Quantity :Packaging quantity.

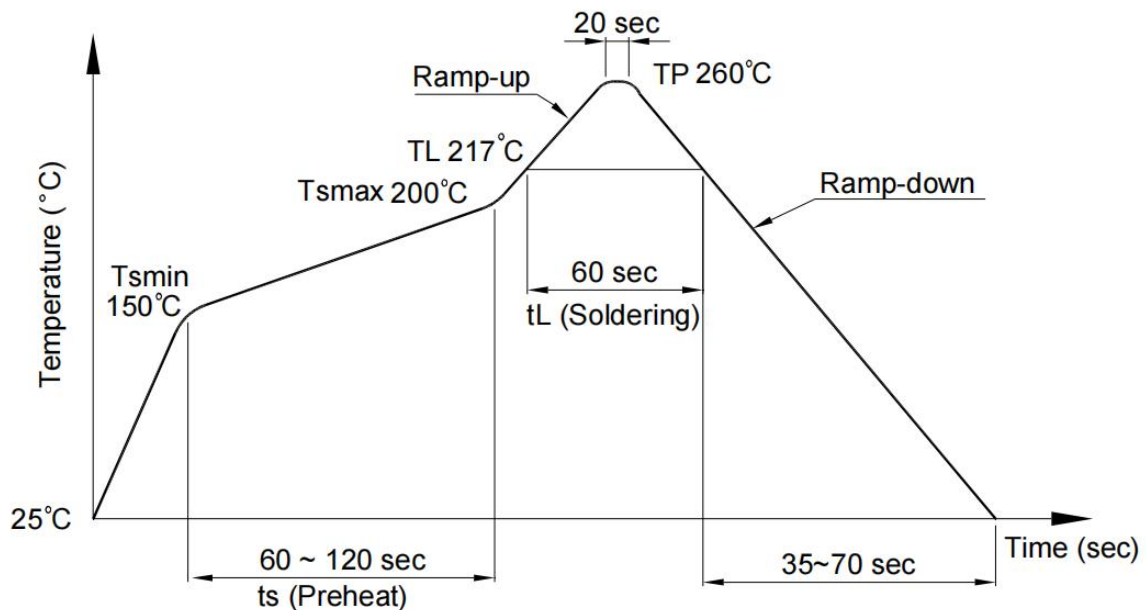
13. Temperature Profile Of Soldering

(1) IR Reflow soldering (JEDEC-STD-020 compliant)

One time soldering reflow is recommended within the condition of temperature and time profile shown below.

Do not solder more than three times

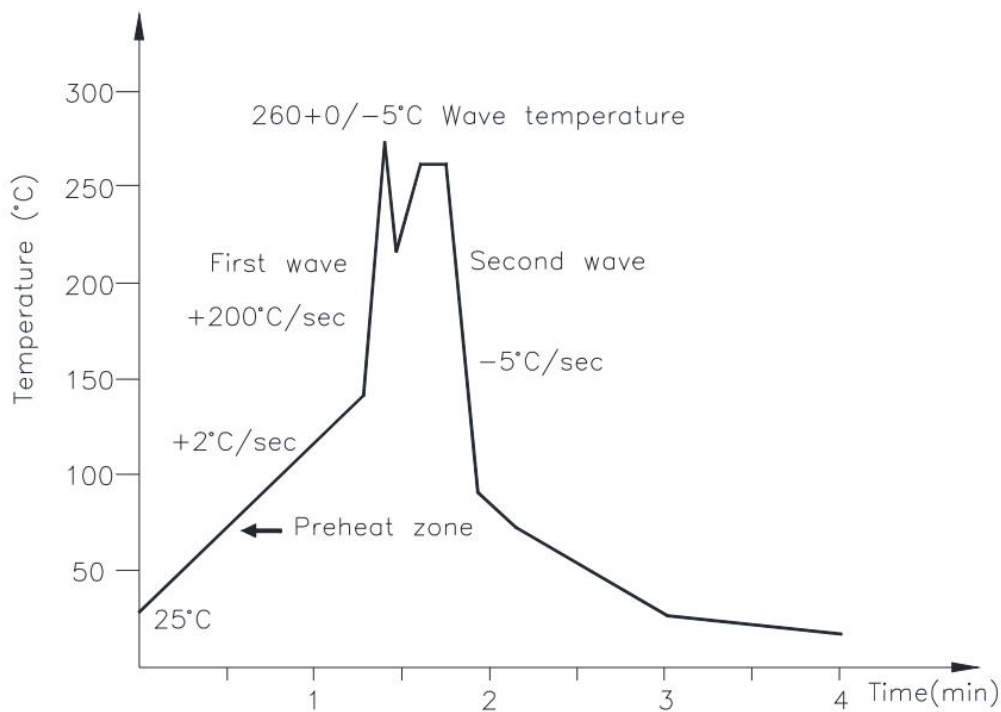
Profile item	Conditions
Preheat	
- Temperature Min (T Smin)	150°C
- Temperature Max (T Smax)	200°C
- Time (min to max) (ts)	90±30 sec
Soldering zone	
- Temperature (TL)	217°C
- Time (t L)	60 sec
Peak Temperature	260°C
Peak Temperature time	20 sec
Ramp-up rate	3°C / sec max.
Ramp-down rate from peak temperature	3~6°C / sec
Reflow times	≤3



(2) Wave soldering (JEDEC22 A111 compliant)

One time soldering is recommended within the condition of temperature.

Temperature	260+0/-5°C
Time	10 sec
Preheat temperature	5 to 140°C
Preheat time	30 to 80 sec



(3) Hand soldering by soldering iron

Allow single lead soldering in every single process. One time soldering is recommended.

Temperature	380+0/-5°C
Time	3 sec max

14. Characteristics Curve

Fig.1 Forward current vs Ambient temperature

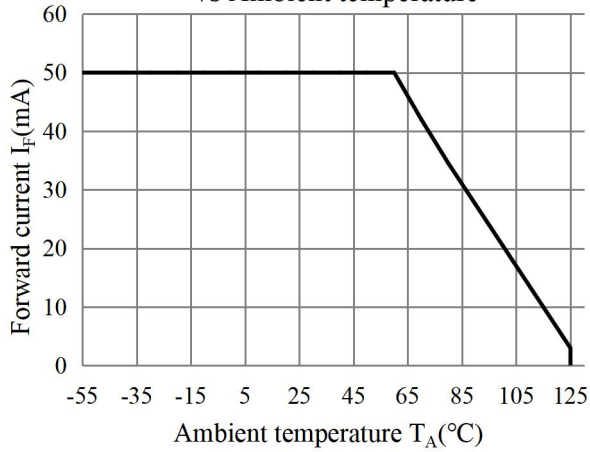


Fig.2 Collector Power Dissipation vs. Ambient temperature

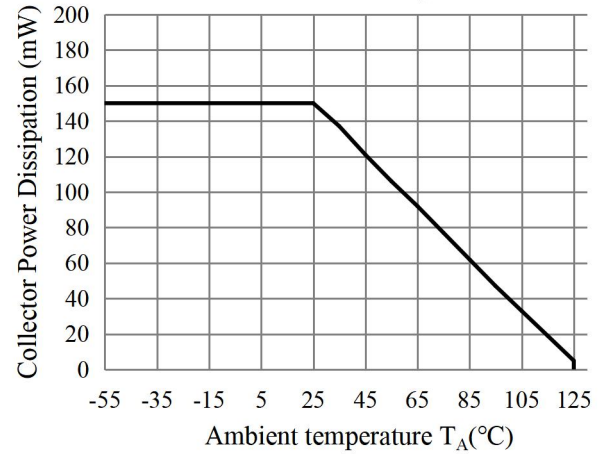


Fig.3 Forward Current vs. Forward Voltage

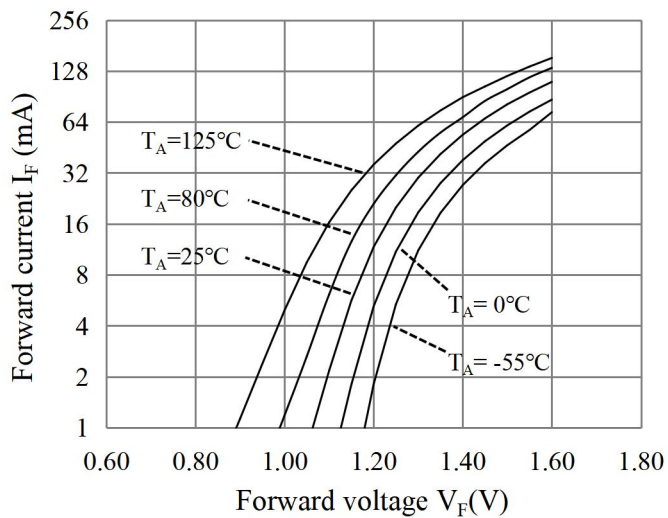


Fig.4 Collector-emitter Saturation Voltage vs. Forward Current

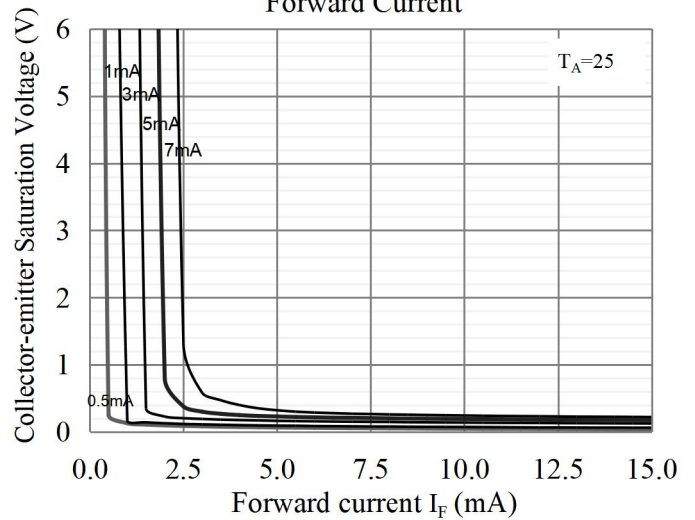


fig.5 Collector Current vs. Non-Saturated Collector Emitter Voltage

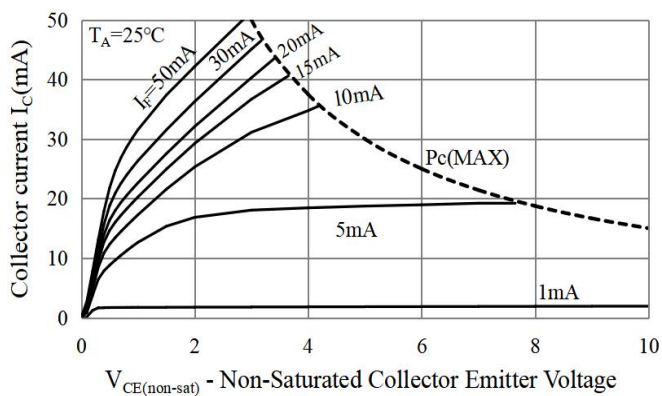


fig.6 Collector Current vs. Non-Saturated Collector Emitter Voltage

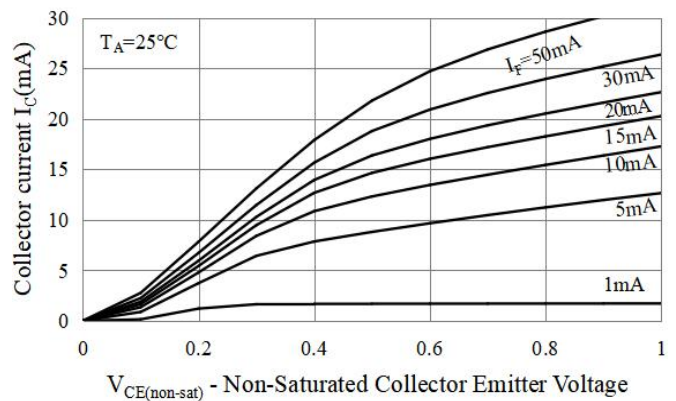


Fig.7 Relative Current Transfer Ratio vs. Ambient Temperature

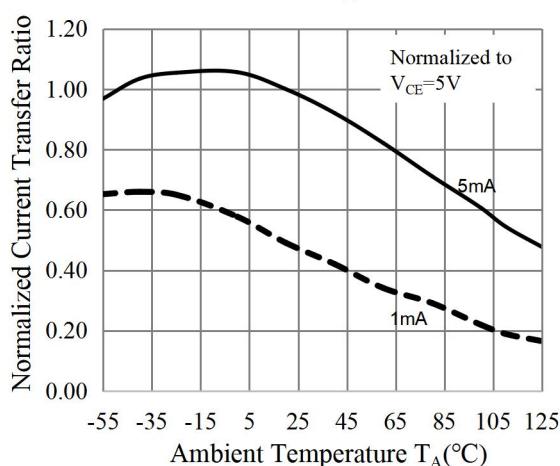


Fig.8 Relative Current Transfer Ratio vs. Ambient Temperature

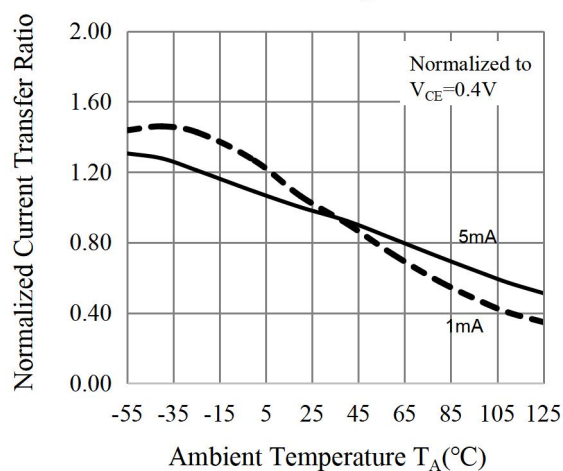


Fig.9 Forward Current vs. Current Transfer Ratio

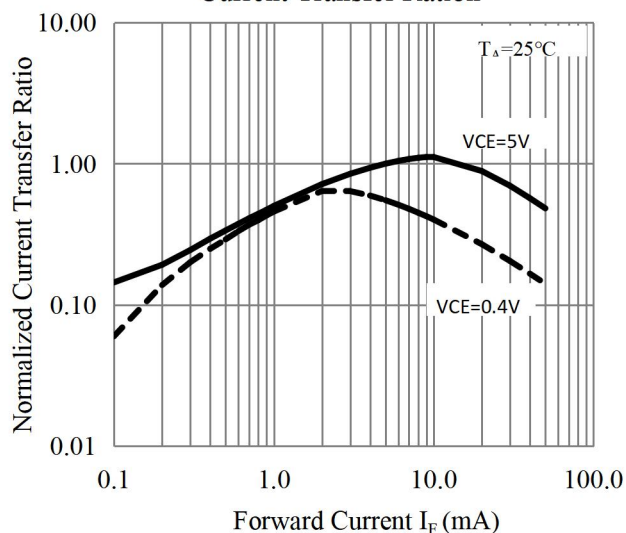


Fig.10 Collector Dark Current vs. Ambient Temperature

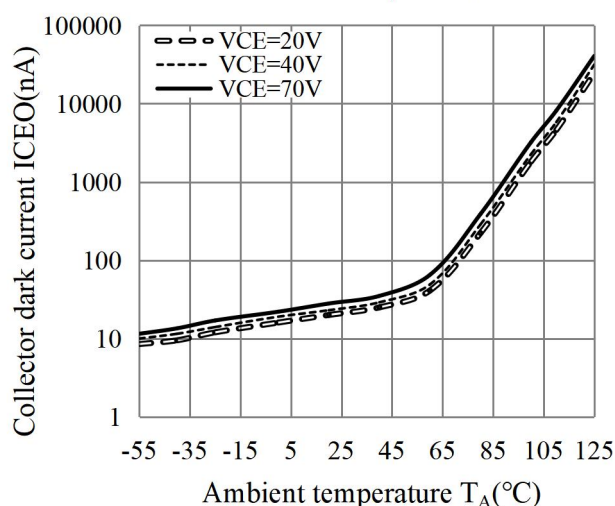


Fig.11 Collector-emitter Saturation Voltage vs. Ambient Temperature

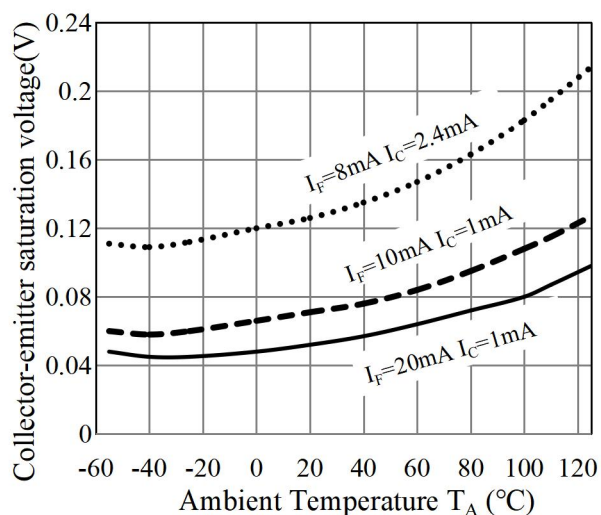


Fig.12 Switching Time vs. Load Resistance

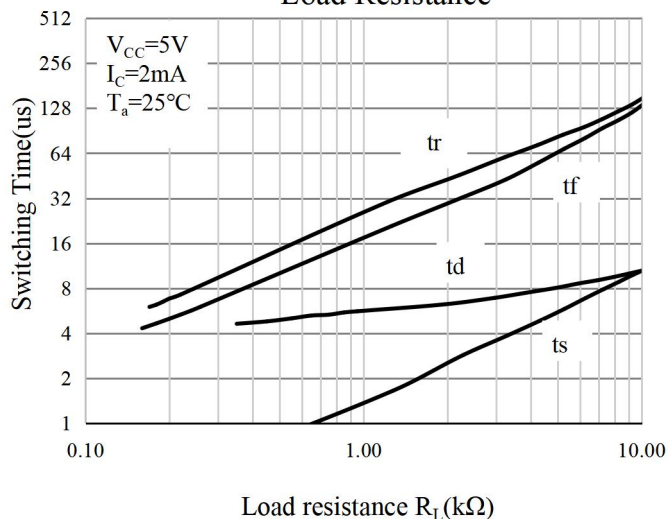


Fig.13 Respinse Time vs. Ambient temperature

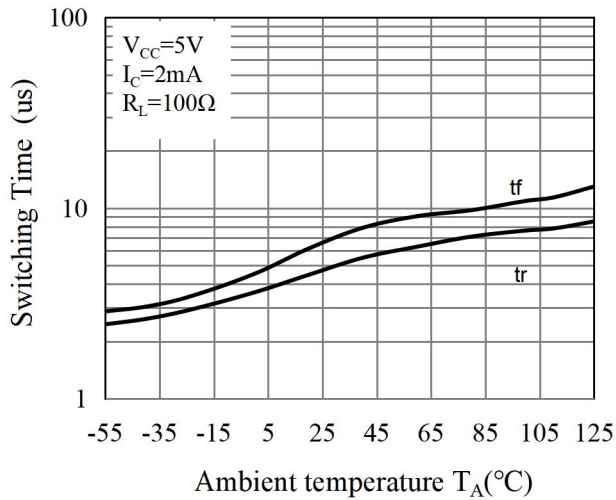
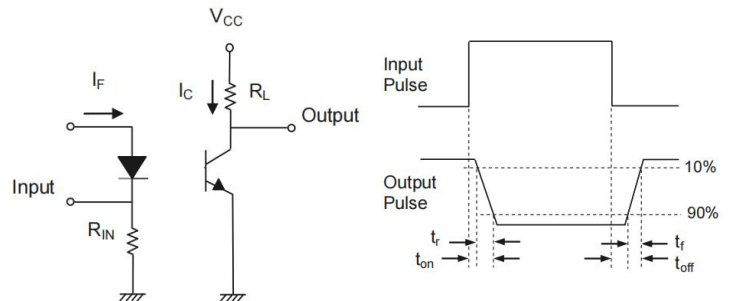


Fig.14 Switching Time Test Circuit & Waveforms



15. NOTES

1. Orient is continually improving the quality, reliability, function or design and Orient reserves the right to make changes without further notices.
2. The products shown in this publication are designed for the general use in electronic applications such as office automation equipment, communications devices, audio/visual equipment, electrical application and instrumentation.
3. For equipment/devices where high reliability or safety is required, such as space applications, nuclear power control equipment, medical equipment, etc, please contact our sales representatives.
4. When requiring a device for any "specific" application, please contact our sales in advice.
5. If there are any questions about the contents of this publication, please contact us at your convenience.
6. The contents described herein are subject to change without prior notice.
7. Immerge unit's body in solder paste is not recommended.